



University of
Nottingham
UK | CHINA | MALAYSIA

School of Mathematical Sciences

A Level 3 Module Spring 2025/26

Time Series Analysis

Instructions:

Time allowed: Three Hours

Answer ALL questions

Candidates may complete the front cover of their answer book and sign their desk card. **DO NOT turn the examination paper over until instructed to do so.**

Only a calculator from approved list A may be used in this examination.

Basic Models	Scientific Calculators	
Aurora HC133	Aurora AX-582	Casio FX82 family
Casio HS-5D	Casio FX83 family	Casio FX85 family
Deli – DL1654	Casio FX350 family	Casio FX570 family
Sharp EL-233	Casio FX991 family	Sharp EL-531 family
	Texas Instruments TI-30 family	Texas BA II+ family

Permitted resources:

Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

Prohibited resources:

All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Appended material: None

Additional material: None

Information for Invigilators: None

Exam Papers must not be removed from the examination venue.

1. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_Z^2 . You should show your working and/or justification clearly in all answers that involve calculations or derivations.

The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = \frac{1}{4}X_{t-1} - \frac{1}{48}X_{t-2} + Z_t$$

- (a) What type of process is $\{X_t : t \in \mathbb{Z}\}$, including its order? [2 marks]
- (b) Is the process $\{X_t : t \in \mathbb{Z}\}$ (weakly) stationary? Justify your answer. [5 marks]
- (c) Write down the Yule-Walker equations for the process $\{X_t : t \in \mathbb{Z}\}$. [1 mark]
- (d) By solving the Yule-Walker equations from (c), determine the autocorrelation function (ACF) of $\{X_t : t \in \mathbb{Z}\}$. [12 marks]
- (e) Hence determine the partial autocorrelation function (PACF) of $\{X_t : t \in \mathbb{Z}\}$. [5 marks]

2. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_Z^2 . You should show your working and/or justification clearly in all answers that involve calculations or derivations.

The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = Z_t + \frac{2}{5}Z_{t-1} + \frac{1}{4}Z_{t-2}.$$

- (a) Determine the autocovariance generating function of $\{X_t : t \in \mathbb{Z}\}$. [4 marks]
- (b) Derive the spectral density function of $\{X_t : t \in \mathbb{Z}\}$, expressing your answer in terms of $\cos(w)$ where $w \in (-\pi, \pi]$ is the frequency. [3 marks]
- (c) Sketch the spectral density function from (b) for values of the frequency, w , where $w \in [0, \pi]$. [5 marks]
- (d) The time series $\{Y_t : t \in \mathbb{Z}\}$ is a linear filtered version of $\{X_t : t \in \mathbb{Z}\}$ where
- $$Y_t = \frac{1}{2}(X_t + X_{t-1}).$$
- Determine the spectral density function of $\{Y_t : t \in \mathbb{Z}\}$. [7 marks]
- (e) Sketch the squared gain for the linear filter described in (d). Hence, describe briefly the effect of this linear filter on the original process $\{X_t : t \in \mathbb{Z}\}$. [6 marks]

3. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_Z^2 . You should show your working and/or justification clearly in all answers that involve calculations or derivations.

The (weakly) stationary process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = 200 - 0.85X_{t-1} + 0.6X_{t-2} + Z_t - 0.25Z_{t-1}.$$

- (a) Determine the mean of $\{X_t : t \in \mathbb{Z}\}$.

[3 marks]

- (b) Hence show that $\{X_t : t \in \mathbb{Z}\}$ can be written in the form

$$X_t - \mu = a(X_{t-1} - \mu) + b(X_{t-2} - \mu) + Z_t - 0.25Z_{t-1}$$

where a , b and μ are real-valued constants.

[2 marks]

- (c) The process $\{X_t : t \in \mathbb{Z}\}$ (considered in (a) and (b)) is assumed to be the annual number of passengers who use a particular railway station (in 1,000s). The following table shows annual numbers of passengers for the years 2022–2025:

Year	2022	2023	2024	2025
No. of passengers (thousands)	159	162	156	168

Using this information, produce an optimal forecast for the number of passengers who use the station for the year 2026. Here, the term 'optimal' implies that the forecasted value is that which minimises the mean square prediction error at the corresponding time point.

[20 marks]

4. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_Z^2 . You should show your working and/or justification clearly in all answers that involve calculations or derivations.

(a) The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = (a + bt)s_t + ct + Z_t$$

where a , b and c are non-zero, real-valued constants and s_t is a seasonal term of period 4, such that $s_t = s_{t-4}$ for all t .

(i) Show that $\{X_t : t \in \mathbb{Z}\}$ is not a stationary process.

[3 marks]

(ii) Suppose that ∇_k is a difference operator that can be applied to a process, where

$$\nabla_k = 1 - B^k$$

for $k \in \mathbb{N}$ and B denotes the backward shift operator.

Using difference operators of this form, show how the process $\{X_t : t \in \mathbb{Z}\}$ can be transformed to a (weakly) stationary process.

[5 marks]

(b) The process $\{U_t : t \in \mathbb{Z}\}$ is a (weakly) stationary process with mean μ , variance σ_u^2 and autocorrelation function $\rho_U(h)$. The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = \frac{1}{2}U_t + \frac{1}{3}Z_{t-1} + Z_t$$

and the processes $\{U_t : t \in \mathbb{Z}\}$ and $\{Z_t : t \in \mathbb{Z}\}$ are independent.

(i) Determine the autocorrelation function of $\{Y_t : t \in \mathbb{Z}\}$, in terms of $\rho_U(h)$ and the process parameters.

[8 marks]

(ii) Is $\{Y_t : t \in \mathbb{Z}\}$ a (weakly) stationary process? Justify your answer.

[4 marks]

(iii) By considering convergence in mean square, show that $\{Y_t : t \in \mathbb{Z}\}$ converges to the MA(1) process

$$Y_t = \frac{\mu}{2} + \frac{1}{3}Z_{t-1} + Z_t$$

as $\sigma_u^2 \rightarrow 0$.

[5 marks]